

Converting ER Model to Relational Tables

Q. How is an ER Diagram mapped into a Relational Database schema? Explain the rules for converting Entities, Attributes, and the various Relationships (1:1, 1:N, M:N) into database tables.

Definition to Remember

ER-to-Tables,

1. Rule 1: Strong Entities and Attributes

* Every **Strong Entity Type** -> becomes a separate **Table**.

* Simp

2. Rule 2: Multi-Valued Attributes

* Cannot store a multi-valued attribute in the main table (relational tables require **atomic** values).

* **So**

3. Rule 3: Weak Entities

* Weak Entity -> converted into a **separate table** with its own attributes.

* Add

4. Rule 4: Mapping Binary Relationships

| Cardinality | New Table Needed? | Action |

| :--- | :-

1:1 Mapping:

MANAC

1:N Mapping:

DEPAR

M:N Mapping:

STUDE

Must-Write Points (for 10 marks)

1. ER-to

Quick Recall

Strong
M:N (Ju